

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour

Total Marks: 50

Annexure A

Scenario Based Assignment

Code of Conduct:

I G. Pallavi (192311208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pallavi

20 + 20 + 25 + 18 + 13
(96) 14

1. Scenario

A marine research team deploys an autonomous underwater vehicle (AUV) to locate survivors near a sunken vessel. The environment is highly dynamic - underwater currents shift debris and visibility, and travel cost changes with depth and current strength. The AUV has access to: a heuristic estimate (straight-line distance to the last known survivor location), partial real-time sonar updates about newly obstructed paths, and variable movement costs depending on current strength and depth. Given the urgency of survivor rescue, the AUV must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic.
- iii. Analyze the impact of heuristic accuracy on both algorithms.
- iv. Discuss how dynamic changes (shifting debris and visibility) affect search performance.
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

2. Scenario

A cloud computing company implements an AI system to balance workloads across thousands of servers in its data centers. The objective is to minimize energy consumption, response latency, and hardware wear. The system must continuously redistribute tasks based on real-time server load. However, the search space is extremely large, incoming request patterns change dynamically, and the system may get stuck in locally optimal load distributions.

Tasks:

Formulate this as an optimization problem and explain why traditional search is inefficient.

- i. Explain how Hill Climbing would work in this scenario and identify its limitations.
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.
- iv. Recommend the best approach and justify your choice based on scalability and adaptability.

3. Scenario

A search-and-rescue drone is deployed inside a building that has partially collapsed after an earthquake. It must: navigate safely to locate survivors, avoid hazards (unstable debris, gas leaks), operate with limited sensing range, and make decisions without a complete structural map. The drone updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search.
- ii. Analyze the challenges of partial observability and dynamic environments.
- iii. Describe how the drone builds and updates its internal model.
- iv. Compare online search with uninformed and informed search strategies.
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

Autonomous Underwater vehicle (AUV) - Search and Rescue:-
Scenario:-

A marine research team uses an Autonomous underwater vehicle (AUV) to locate survivors near a sunken vessel. The underwater environment is dynamic because currents change, debris shifts, visibility varies, and movement costs depend on depth and current strength.

(i) Greedy Best-First Search:-

Greedy Best-First Search (GBFS) selects the node that appears closest to the goal according to a heuristic function.

The evaluation function is :

$$f(n) = h(n)$$

where $h(n)$ is the estimated distance from the current position to the last known survivor location.

In this scenario, the AUV would continuously choose the path that appears to bring it closest the survivor.

Strengths :-

- Fast when the heuristic is accurate.
- Can quickly move toward the estimated survivors location.
- Requires less computation than algorithms that consider complete path costs.
- Useful when rescue speed is more important than guaranteed optimality.

Weakness :-

- It ignores the cost already spent reaching a node.
- It may select a short but dangerous route with strong currents or unstable debris.
- An inaccurate survivor location can mislead the search.
- It can get stuck exploring an apparently promising route that is actually blocked.
- It does not guarantee the minimum-cost or

Safest path.

∴ GBFS is fast but not necessarily optimal or safe in a highly uncertain underwater environment.

Search :-

A* combines the actual cost of reaching a node with the estimated cost to the goal.

Its evaluation function is :

$$f(n) = g(n) + h(n)$$

where :

- $g(n)$ = actual cost from the starting point to node n .
- $h(n)$ = estimated cost from n to the solution.
- $f(n)$ = estimated total cost of the solution through n .

How A* balances cost and heuristic :

- GBFS : considers only $h(n)$.
- A* considers both $g(n)$ and $h(n)$.

(iii) Impact of Heuristic Accuracy :-

The quality of the heuristic has a major effect on both algorithms.

Greedy Best-First Search :

GBFS depends heavily on the heuristic because :

$$f(n) = h(n)$$

- ⇒ Accurate heuristic → faster and better decisions.
- ⇒ Inaccurate heuristic → misleading decisions.

A^* is less independent on the heuristic because it also considers the actual cost.
 $f(n) = g(n) + h(n)$

- (iv) Dynamic Changes :-
Shifting debris, changing currents, and poor visibility can make existing paths invalid. Both algorithms need to update the search and replan when new sonar information arrives.

2. Cloud workload Optimization :-

Problem Formulation :-

The objective is to minimize :

$$\text{cost} = w_1 (\text{Energy}) + w_2 (\text{latency}) + w_3 (\text{Hardware wear})$$

$\Rightarrow$ The system must distribute tasks effectively among thousands of servers.

Why Traditional Search is inefficient :

The number of possible task-server combinations is extremely large.
 $\Rightarrow$ Exhaustive Search is therefore too slow for real-time workload management.

(i) Hill Climbing :-

Hill climbing starts with a workload distribution and repeatedly moves to a better neighbouring solution.

Local Maxima, Plateaus and Ridges :-

⇒ Local maximum : Good nearby solution but not, the global best.

⇒ Plateau : Many solutions have similar values, so there is no clear improvement.

⇒ Ridge : Improvement requires several coordinated moves.

(iii) Simulated Annealing / Genetic Algorithm :-

Simulated Annealing sometimes accepts worse solutions to escape local optima.

Genetic Algorithm uses Selection, Crossover, and mutation to explore many possible workload distributions.

(iv) Recommendation :-

Genetic algorithm is suitable because it can handle a large search space, avoid local optima, and adapt to changing workloads.

3. Search - and - Resume Drone :-

(i) Need for online Search :-

The drone does not have a complete map and discovers obstacles and survivors while moving. Therefore, it must search and act.

(ii) Partial Observability and Dynamic Environment:-

The drone has limited sensing and cannot see the entire building. Debris may collapse, gas leaks may spread, and paths may become blocked.

(iii) Internal Model:-

The drone uses sensor data to continuously update its internal map.

Sense $\rightarrow$ Update Map $\rightarrow$ Plan $\rightarrow$ Act

When new information is discovered; it replans the route.

(iv) Comparison:-

$\Rightarrow$ Uninformed Search:- No heuristic; explores without guidance.

$\Rightarrow$ Informed Search:- Uses heuristics such as distance to survivor.

$\Rightarrow$ Online Search:- Learns about the environment while operating.

(v) Recommendation:-

Use an online heuristic-guided search with continuous replanning. It provides; Adaptability, Safety, Real-time decision making.